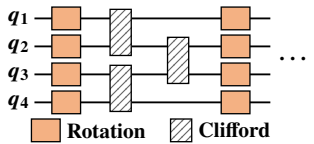
Goal

Quantum Circuit



127 Qubits, 20 Steps

Expectation Value

One Number: the Average Result of a Measurement

Two Ways to Compute It

Method	Stores	Limit
State Vector	2^n Amplitudes	About 50 Qubits
Pauli Propagation	A List of Terms	Number of Terms

Pauli Propagation: Every Gate Does Three Things

Branch ❶

Coefficient	Pauli String
1.0	Z

Rotation Gate

$\cos\theta$	Z
$\sin\theta$	Y

One Term Becomes Two

Merge ❷

Coefficient	Pauli String
0.20	X Z
0.15	X Z

Same Pauli String

0.35	X Z

Equal Strings Are Added

Truncate ❸

Coefficient	Pauli String
0.35	X Z
0.12	Y I
0.004	Z X

Cutoff = 0.01

✓	0.35	X Z
✓	0.12	Y I
✗	0.004	Z X

Small Terms Are Dropped

A Term = Coefficient × Pauli String

The List Grows

127-Qubit Kicked Ising

Cutoff	Terms Kept	Size
1e-4	22.5 Million	0.9 GB
1e-5	1.4 Billion	56 GB
1e-6	100.1 Billion	4.0 TB

One GPU (40 GB) Holds About 100 Million Terms

More Terms Kept, More Accurate

30-Qubit Circuit With a Known Answer

Cutoff	Error
1e-3	0.019
1e-7	0.00007

Lower Cutoff → More Terms Kept → More Accurate Answer → More Memory and Time: Many GPUs and Nodes Are Needed